PROGRAM 3: Question-Answering System Using a Pretrained LLM

AIM

To integrate a pretrained language model with Python to implement a question-answering system.

ALGORITHM

1. Import the Transformers library.
2. Load a pretrained question-answering model.
3. Define a reference context.
4. Accept a question.
5. Extract the answer from the context.
6. Display the answer.

CODE

```
from transformers import pipeline


qa = pipeline(
    "question-answering",
    model="distilbert-base-cased-distilled-squad"
)


context = """
Artificial Intelligence is a branch of computer science.
Machine Learning is a subset of Artificial Intelligence.
Deep Learning uses neural networks.
"""


question = input("Enter your question: ")


result = qa(
    question=question,
    context=context
)
```

print("\nAnswer:")

print(result["answer"])

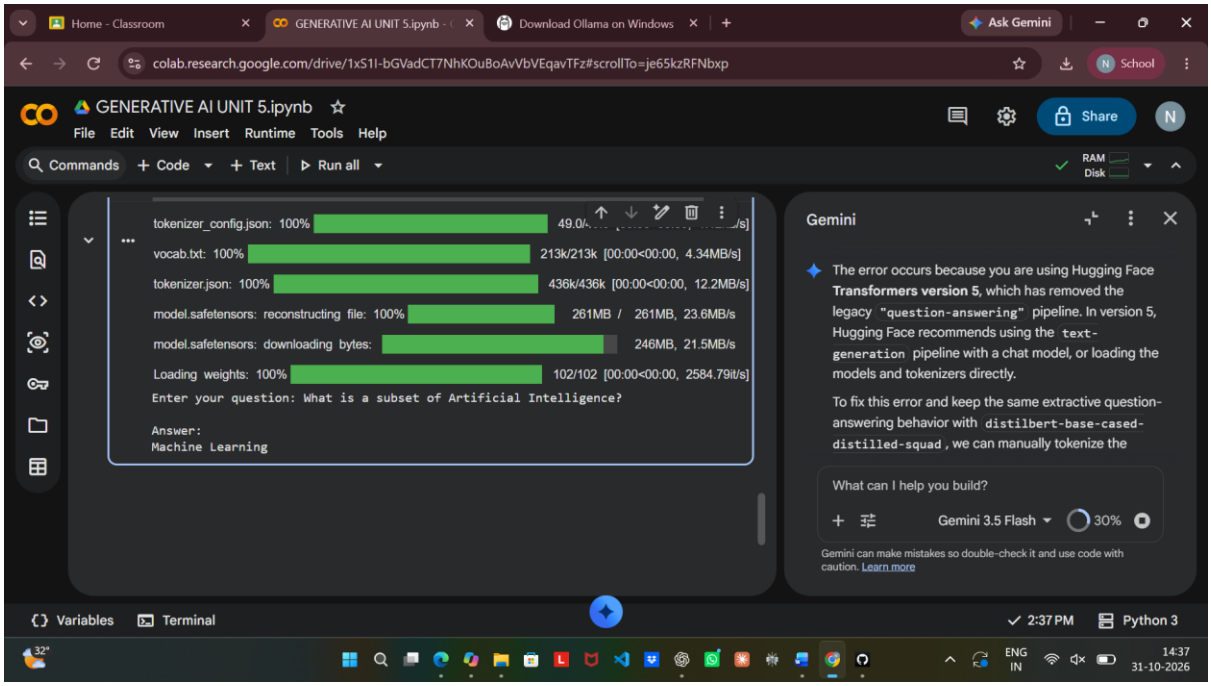
INPUT

Enter your question: What is a subset of Artificial Intelligence?

OUTPUT

Answer:

Machine Learning



RESULT

Thus, a question-answering system was implemented using a pretrained language model.